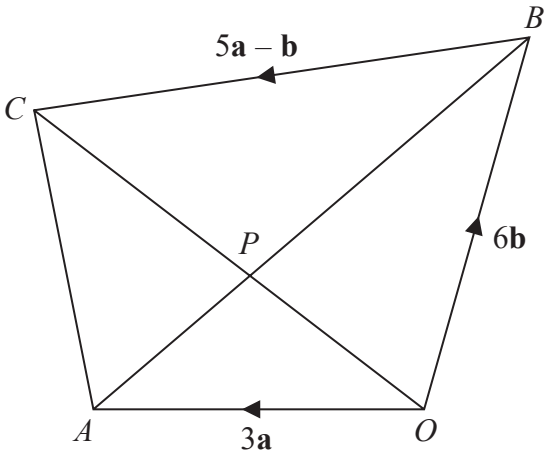
21 $OACB$ is a quadrilateral.
APB and OPC are straight lines.



$\overrightarrow{OA} = 3\mathbf{a}$ $\overrightarrow{OB} = 6\mathbf{b}$ $\overrightarrow{BC} = 5\mathbf{a} - \mathbf{b}$

$OP : PC = 2 : 3$

Work out the ratio $AP : PB$
You must show all your working.

$OC = OB + BC = 6b + (5a - b) = 5a + 5b$

1 mark
Add vectors along a connected route.

$OP = 2/5$ of $OC = 2a + 2b$

$AB = OB - OA = 6b - 3a$

1 mark
Use 2/(2 + 3), then subtract position vectors.

$AP = OP - OA = (2a + 2b) - 3a = -a + 2b$

$PB = OB - OP = 6b - (2a + 2b) = -2a + 4b$

1 mark
Obtain two vectors along line AB.

$PB = 2AP$, so $AP : PB = 1 : 2$

1 mark
Their coefficients have the same scale factor 2.

1 mark
AP is one part and PB is two parts.

1 : 2

(Total for Question 21 is 5 marks)

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